Clinical Analytics Report

Technical Project Documentation & Strategic Pivot Assessment

TARGET DATABASE	PRIMARY TABLE	EXECUTION STATUS
Healthcare Analytics	**healthcare_dataset_cleaned**	**Pivoted & Completed**

1. Executive Summary & Original Objective

The initial scope of this data analytics initiative was framed around tracking and evaluating **30-day hospital readmission rates**. In healthcare analytics, 30-day readmissions serve as a critical Key Performance Indicator (KPI) for measuring hospital care quality, treatment efficacy, and post-discharge patient stability. To achieve this, our primary technical strategy relied on deploying an SQL self-join architecture. By matching a primary patient record discharge date (`t1`) against a subsequent visit date (`t2`) for the exact same `Patient_ID` within a 1 to 30-day window, we aimed to uncover readmission patterns and evaluate risk factors.

> **Initial Technical Impediment**
> Executing self-joins and aggregated uniqueness checks repeatedly yielded zero rows. Diagnostic validation confirmed that every `Patient_ID` in the target table appeared precisely once, completely lacking longitudinal encounter tracking data.

2. Architectural Deviation & Schema Reality

A thorough inspection of the database schema via SQL Server Management Studio (SSMS) revealed that `dbo.healthcare_dataset_cleaned` is structured strictly as a **cross-sectional patient registry** rather than a longitudinal encounter database. Each record encapsulates a single health snapshot containing immutable biometrics (`BMI`, `Blood_Pressure`, `Cholesterol_Level`, `Heart_Rate`), a primary `Diagnosis`, and an assigned `Treatment_Plan`, alongside a single `Follow_Up_Date`.

Attempting to force a 30-day readmission metric onto a cross-sectional data model would introduce fundamental statistical errors. Consequently, the project underwent a professional analytic pivot: transitioning from longitudinal tracking to a comprehensive **Patient Risk & Biometric Profiling Model**.

3. Pivot Execution: Biometric Risk & Demographic Insights

To maximize the clinical value of the available dataset, our analysis shifted toward evaluating how lifestyle markers, vitals, and demographics correlate across different primary medical diagnoses. Four core aggregations were established.

Primary Clinical Risk Summary

Diagnosis	Total Patients	Avg Age	Avg BMI	Avg Cholesterol	Avg Heart Rate
Healthy	104	53.6	30.4	230.3	80.4
Hyperlipidemia	101	48.9	29.5	224.0	81.1
Hypertension	101	49.0	29.2	234.2	79.4
Coronary Artery Disease	101	49.3	28.2	226.9	79.6
Diabetes	93	43.0	29.8	212.4	78.8

4. Key Strategic Takeaways & Next Steps

The pivoted analysis successfully extracted actionable intelligence despite initial structural limitations. Key observations indicate that conditions like Hypertension and Coronary Artery Disease maintain elevated average cholesterol thresholds above 225 mg/dL, while Diabetes presents in a notably younger average cohort (Age 43.0) compared to healthy baselines.

> **Readiness for Power BI Visualizations**
> All aggregate datasets have been validated and structured in SSMS. These clean outputs provide the exact foundation required for building executive dashboards in Power BI.

Report Conclusion: Documenting database constraints and executing a clean methodological pivot preserves data integrity, transforming a potential query failure into a rigorous cross-sectional health profiling study.